

Evaluating temporal sentiment as a signal for momentum-like scoreline swings in CS2 match threads

Course: DS680 — NLP final project

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Abstract

We ask whether **time-binned sentiment velocity** from audience comments aligns with **heuristic scoreline swing markers** in a **self-built** SQLite corpus of HLTV-style Counter-Strike 2 match threads (timestamped lines with embedded score contexts). We compare **Multinomial Naive Bayes** (unigram and bigram) to **embedding + LSTM** classifiers that score each comment from its token sequence, then map predicted probabilities to a scalar score and compute velocity over fixed-width bins. Training and evaluation use **match-level splits** and both **weak lexicon labels** and a **hand-labeled gold subset**. A second LSTM configuration (**hidden size 64** vs **128**) satisfies the course expectation for hyperparameter/architecture exploration. On 2,174 comments over 16 matches, the larger LSTM achieves the best **macro-F1** on held-out fixtures under weak supervision, but **gold** evaluation remains difficult for the minority negative class. **Aggregate** correlations between per-comment scores and our swing proxy are **near zero**; we treat that as the primary empirical finding rather than evidence of reliable momentum prediction.

1. Introduction

Live match threads are short, slang-heavy, and only loosely tied to latent game state. We do **not** claim to recover full momentum from text alone; we **evaluate** a simple two-stage pipeline—(i) sentiment classifiers, (ii) temporal aggregation and correlation against a transparent proxy derived from `score_context` strings—and report where it succeeds and fails.

2. Related work

[1] Moon et al. (EMNLP 2023) study **norm violations in Twitch live chat**, show that models trained on forum-style data degrade, and improve moderation with **human-identified contextual features** [1]. *Limitation*: task is violation detection, not continuous polarity derivatives; data are English Twitch, not CS2 forums. *Gap*: we target **score-linked sentiment dynamics**, not moderation labels.

[2] Gao et al. (ACL 2023) introduce **LiveChat**, a large **Chinese** live-streaming dialogue benchmark emphasizing addressee and response modeling [2]. *Limitation*: language and task differ from ours; models are evaluated on dialogue generation/retrieval, not scoreline-linked velocity. *Gap*: our corpus is smaller, English, and **paired with partial scorelines** per comment.

[3] Wu et al. (EMNLP 2023) propose **SentiStream**, co-training for **sentiment in evolving streams** [3]. *Limitation*: streams are generic social feeds, not esports; method assumes ongoing label adaptation rather than a fixed match window. *Gap*: we use **static weak + gold labels** and explicit **in-match** time bins rather than open-ended stream drift.

[4] Chen et al. (EMNLP 2024) present **D2R**, a **multimodal** routing network for sentiment detection [4]. *Limitation*: requires aligned non-text modalities; our setting is **text-only**. *Gap*: highlights that **fusion and routing** matter for affect; we isolate **lexical temporal** signal only.

[5] Shukla et al. (ACL 2025) audit **Twitch AutoMod**, finding high bypass and false-positive rates, underscoring **context sensitivity** of platform classifiers [5]. *Limitation*: hate-speech audit, not sentiment velocity. *Gap*: motivates caution when repurposing moderation-style features for analytics on similar chat.

3. Methodology

Corpus. We assembled comments in `data/hltv_sentiment.db` from **publicly accessible** HLTV-style sources using the project collector; this is **not** a standard packaged NLP dataset. Ingest respects site

terms and **robots.txt** (automated forum fetch off by default); see `docs/HLTV_SENTIMENT_COLLECTION.md` for compliance, import/JSONL options, and what we redistribute (code, metrics, aggregates—**not** necessarily raw text dumps).

Labels and splits. Weak labels use `nlp.preprocess + nlp.weak_labels`; gold integers in $\{0,1,2\}$ can be merged from CSV. Splits are **by_match** with 15% validation and 15% test so rows from one `match_id` do not cross partitions. Optional **pre/during/post** phases use `nlp.time_windows`; momentum analysis uses **during** comments.

Models. **NB:** CountVectorizer + MultinomialNB (`scripts/train_sentiment_nb.py`). **LSTM:** CommentLSTM embeds **word tokens within each comment** (max length 64), mean-pools LSTM outputs, and classifies; it does **not** encode the full ordered list of comments as one sequence—**cross-comment time** enters only when we bin scores (`nlp.velocity.velocity_per_bin`). **Swing proxy:** `swing_labels_from_context` on parsed round differentials in `score_context`; **lags** are comment-index offsets (`scripts/eval_sentiment_momentum.py`).

Reproduction. Full command sequence and PDF build: `nlp/ProjectDocs/REPRO_COMMANDS.md`.

4. Experiments

We train weak-supervised baselines with seed **42**, evaluate on the held-out **match split** (`sentiment_models/**/_metrics.json` and `sentiment_eval/metrics_*.json`), and run momentum JSON export for the **default (hidden 128)** LSTM and NB (qualitative pattern matches the smaller LSTM).

5. Results

5.1 Dataset snapshot

From `dataset_stats.json`: **2,174** comments, **16** matches, **530** gold-labeled rows; weak counts neg/neu/pos **228 / 1,618 / 328**; timestamps and score contexts **100%** populated in this build.

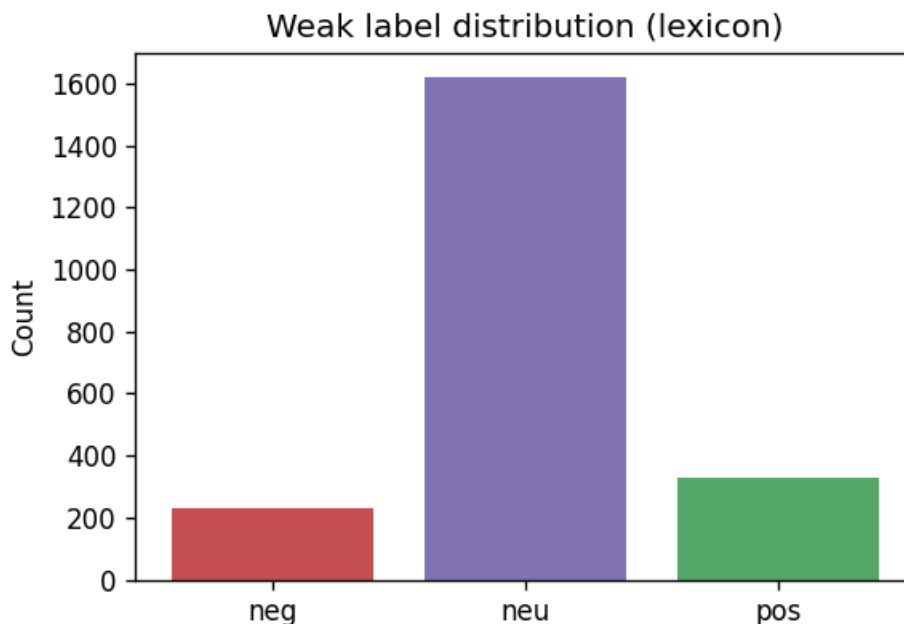


Figure 1: Weak-label distribution

Additional histograms (comments per match, phase counts) remain in `nlp/ProjectDocs/figures/` for inspection.

5.2 Classification (weak labels, test split, n=390)

Per-class metrics from `sklearn` classification reports on the held-out match split (NB unigram and both LSTMs: `sentiment_eval/metrics_*.json`; NB bigram: `sentiment_models/nb_bigram/nb_bigram_metrics.json` test split—same protocol as unigram).

Macro-F1 and negative class (minority, hardest).

Model	Macro-F1	Neg P	Neg R	Neg F1
NB unigram	0.606	0.55	0.32	0.40
NB bigram	0.596	0.80	0.21	0.33
LSTM, hidden 128	0.658	0.48	0.42	0.45
LSTM, hidden 64	0.640	0.83	0.26	0.40

Neutral and positive classes.

Model	Neu P	Neu R	Neu F1	Pos P	Pos R	Pos F1
NB unigram	0.85	0.94	0.89	0.61	0.46	0.53
NB bigram	0.82	0.97	0.89	0.73	0.46	0.57
LSTM, hidden 128	0.89	0.92	0.91	0.64	0.60	0.62
LSTM, hidden 64	0.86	0.94	0.90	0.62	0.62	0.62

The **negative** class remains the weakest category. Reporting precision and recall separately shows that the difficulty is driven primarily by **limited recovery of negative examples** (low **recall** on neg for unigram and bigram NB, and for the small LSTM despite high neg precision on weak labels—i.e. **missed negatives** under class imbalance) rather than headline accuracy alone.

Bigram NB underperforms unigram on **macro-F1** despite slightly higher positive F1: with **short messages** and a **match-aware** split, many bigrams are **sparse** or appear only in one fixture, so the vectorizer sees **fragmented statistics** relative to unigrams; the model can also **overfit** bigram patterns on the training matches without transferring.

LSTM variants: reducing hidden size from 128 to 64 **lowers macro-F1** on weak labels. On **gold** labels (n=101 test rows with human annotations), the default LSTM retains macro-F1 **0.61** vs **0.49** for the 64-unit model—the smaller net shows **zero precision/recall on negative** in this split, illustrating **variance** under class imbalance.

5.3 Momentum proxy

Figure: mean lag correlation (Pearson / Spearman) between per-comment sentiment scores and the swing proxy, aggregated across matches (**default LSTM, during** phase).

With **during**-phase filtering, **five** matches contribute to aggregate momentum statistics. Mean lag correlations (Pearson / Spearman) stay **close to zero** across lag offsets (see `momentum_report_lstm.json`); we interpret this as **weak linear coupling** between chat polarity scores and our coarse proxy—not as proof of absence of all structure (see exemplar plots in `figures/`).

6. Discussion

Weak-label evaluation **overstates** agreement with the lexicon; **gold** metrics are the better deployment proxy. The swing label is a **heuristic** on partial score strings, not a full demo tick stream. For **stronger** match-level

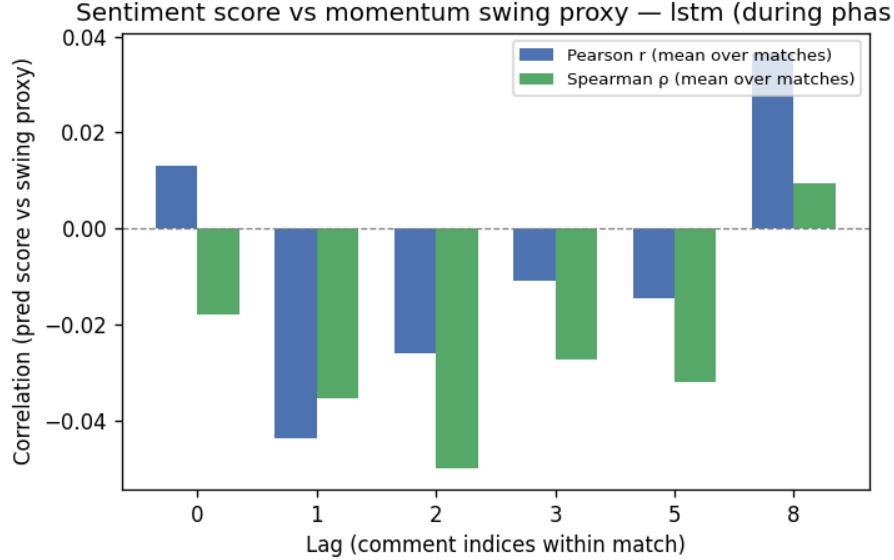


Figure 2: Mean lag correlations (LSTM scores, default model)

modeling, a hierarchical encoder over comment sequences would be more faithful than per-comment LSTMs plus post-hoc binning.

7. Conclusion

We documented an honest end-to-end pipeline: match-aware sentiment baselines, an extra LSTM width ablation, and velocity-based analysis against a scoreline proxy. The headline result is **cautious: macro-F1** gains from the larger LSTM are real on weak labels, but **gold** performance and **momentum correlations** show that **forum text alone is a weak stand-in for momentum** at our scale.

References

- [1] J. Moon *et al.*, “Analyzing Norm Violations in Live-Stream Chat,” EMNLP, 2023. <https://aclanthology.org/2023.emnlp-main.55/>
- [2] J. Gao *et al.*, “LiveChat: A Large-Scale Personalized Dialogue Dataset Automatically Constructed from Live Streaming,” ACL, 2023. <https://aclanthology.org/2023.acl-long.858/>
- [3] Y. Wu *et al.*, “SentiStream: A Co-Training Framework for Adaptive Online Sentiment Analysis in Evolving Data Streams,” EMNLP, 2023. <https://aclanthology.org/2023.emnlp-main.380/>
- [4] Y. Chen *et al.*, “D2R: Dual-Branch Dynamic Routing Network for Multimodal Sentiment Detection,” EMNLP, 2024. <https://aclanthology.org/2024.emnlp-main.207/>
- [5] P. Shukla *et al.*, “Silencing Empowerment, Allowing Bigotry: Auditing the Moderation of Hate Speech on Twitch,” ACL, 2025. <https://aclanthology.org/2025.acl-long.1110/>

BibTeX: `references.bib`.